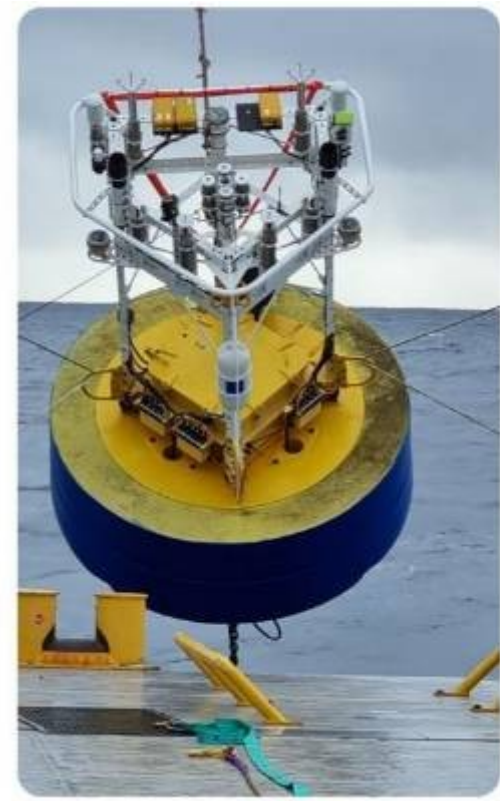


A Deep Dive into IMOS AODN Data

a data workshop
Sunday 30 August 2026



Acknowledgement of Country

IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work

We recognise their role as Australia's first marine scientists and carers of Sea Country.

We pay our respect to Aboriginal and Torres Strait Island peoples past and present.





A Deep Dive into IMOS/AODN data



Time	Session	Speakers
09:00	ARRIVAL & REGISTRATION	
09:30	Welcome & acknowledgement of Country	Eduardo Klein
09:45	IMOS program overview	Eduardo Klein
10:00	Where can I access IMOS data? (overview of architecture: new portal launch, THREDDS, S3)	Eduardo Klein
10:45	MORNING TEA	
11:00	What is Cloud-Optimised Data?	Tom Galindo
11:15	AODN Cloud optimise Library	Laurent Besnard
12:30	Working with IMOS cloud optimised datasets	Laurent Besnard, Tom Galindo, Denisse Fierro
13:00	LUNCH	
13:30	Guided tutorial: Using IMOS data in R, Python AODN CO tools	Laurent Besnard, Tom Galindo, Denisse Fierro
14:30	Excercise: definition of case studies and working groups	Facilitators and Participants
15:00	free flowing AFTERNOON TEA	
15:00	Excercise: cont.	Facilitators and Participants
16:00	Results and WORKSHOP CLOSE	Facilitators and Participants

Australia's Integrated Marine Observing System (IMOS):

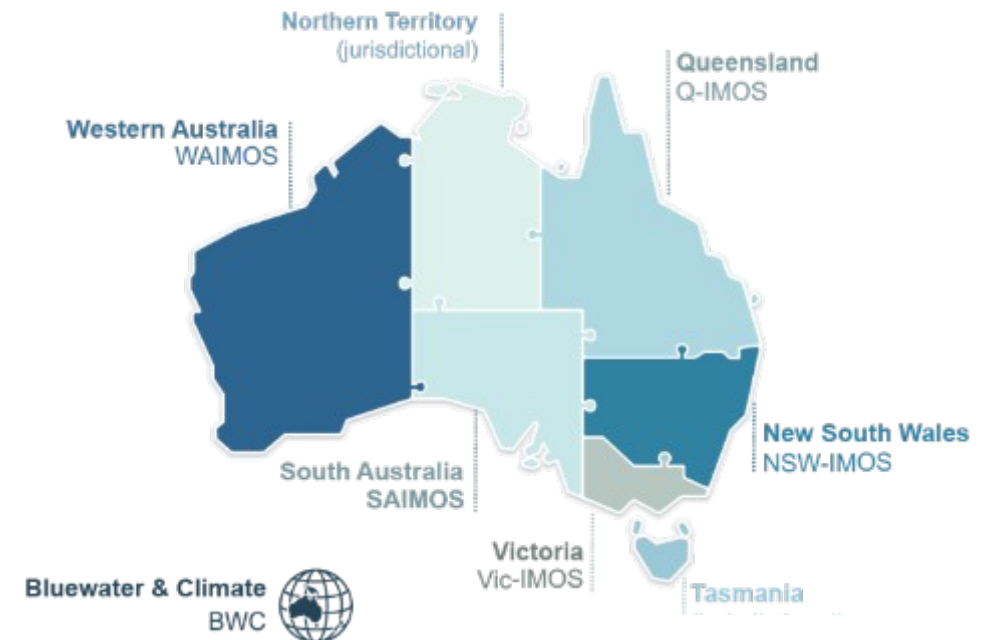
Supporting best practice ocean science, decision-making and industry operations through sustained, high-quality and open data



IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

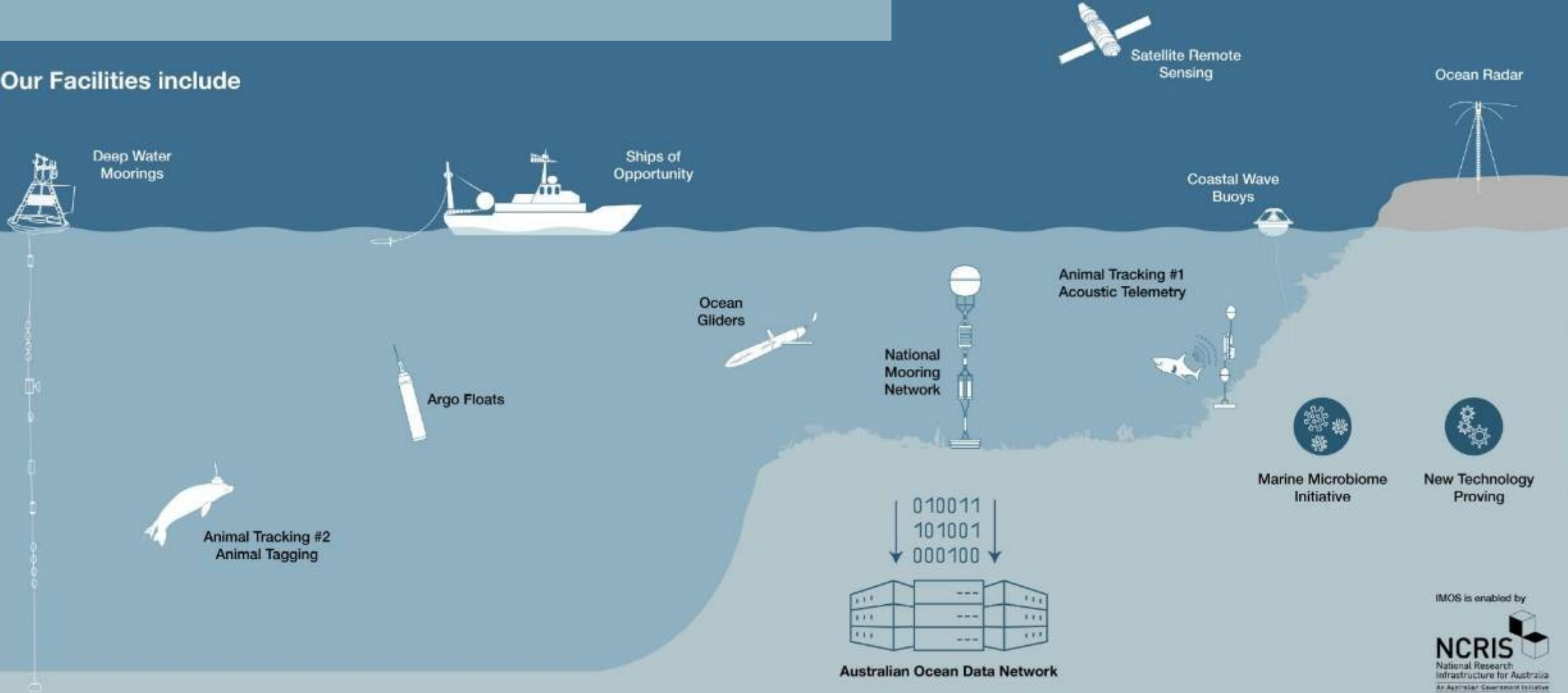
Australia's Integrated Marine Observing System

- A **national, collaborative, research infrastructure** funded by the Australian Government in 2007 to provide **systematic, sustained and scientifically-robust observations of the marine environment** for research, management and industry.
- **IMOS** is guided by community-driven science and implementation planning, organised through jurisdictional Nodes that represent stakeholder needs and underpin what we need to observe and where.



- Platform or Technology-based capability portfolio
- 14 Facilities, 52 sub-Facilities
- IMOS data are collected according to Best Practices in the field
- IMOS data are FAIR (Findable, Accessible, Interoperable, Reusable)

Our Facilities include



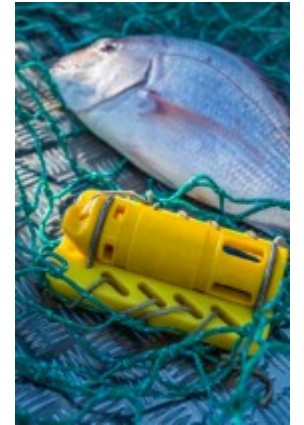
Ships of Opportunity

IMOS operates various sensors onboard a range of SOOPs to observe the ocean along their routes, from the poles to the tropics, and from coastal waters to the deep ocean.

Research Vessels



Fishing Vessels

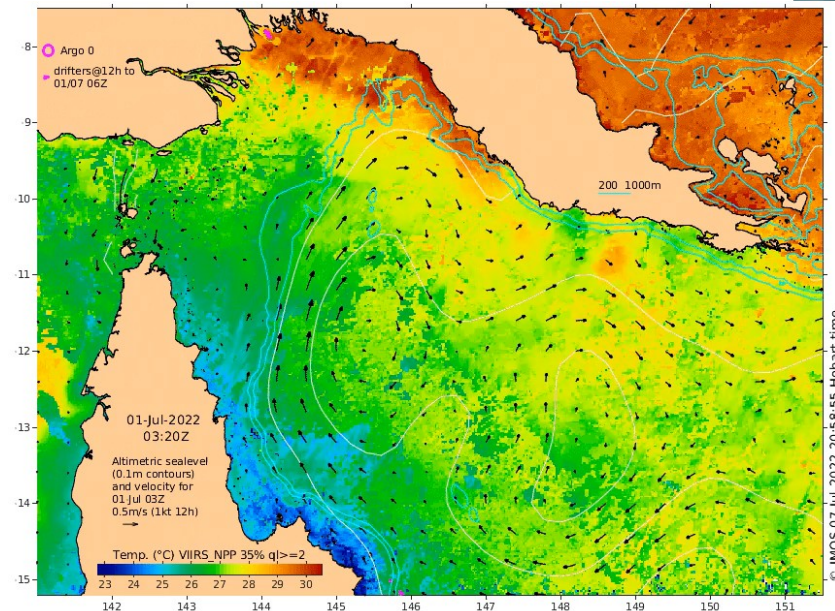


Industry Vessels



Satellite Remote Sensing

IMOS collaborates with various space agencies around the world to collect time series of *in situ* ocean temperature, colour and altimetry to validate and deliver calibrated satellite imagery.



Moored Instruments

IMOS also produces some of the longest ocean data time series in the Southern Hemisphere by deploying oceanographic moorings in coastal, shelf and offshore waters.

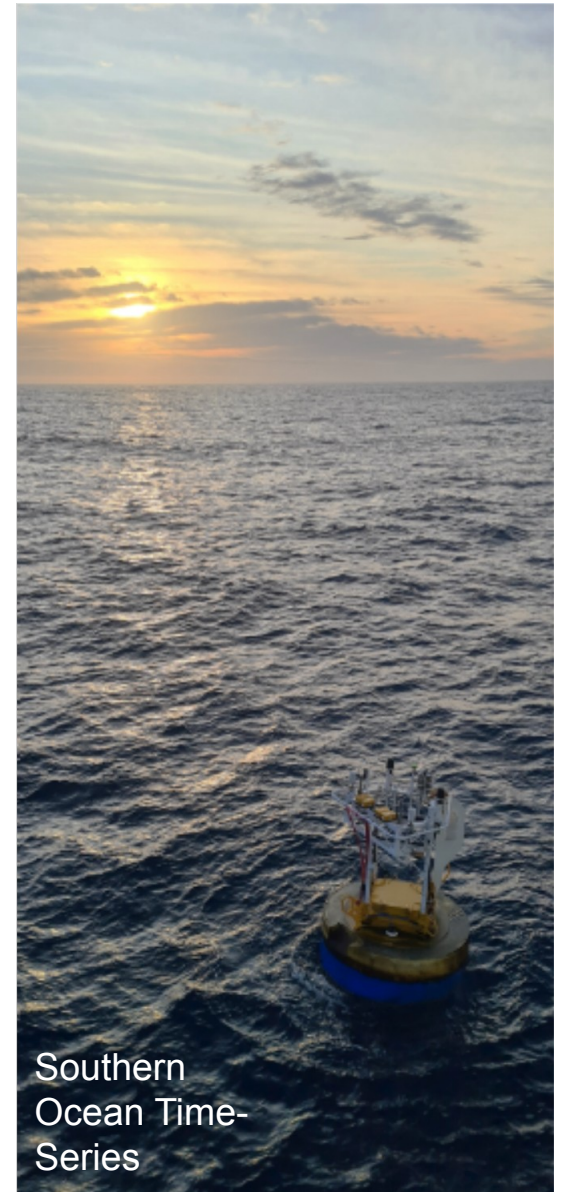
National References
Stations



Coastal Waves
Buoys



Southern
Ocean Time-
Series

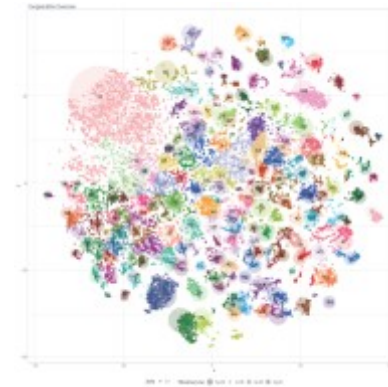


Water Sampling

IMOS conducts regular water sampling to understand patterns of ocean productivity, health and pollution.



Plankton



Nutrients, Microbes
& eDNA

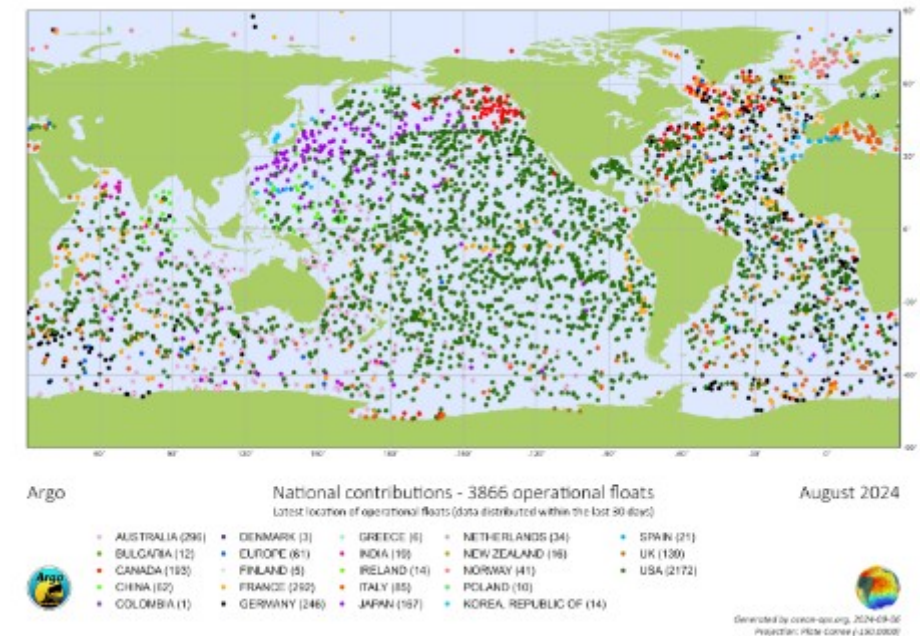


Microplastics



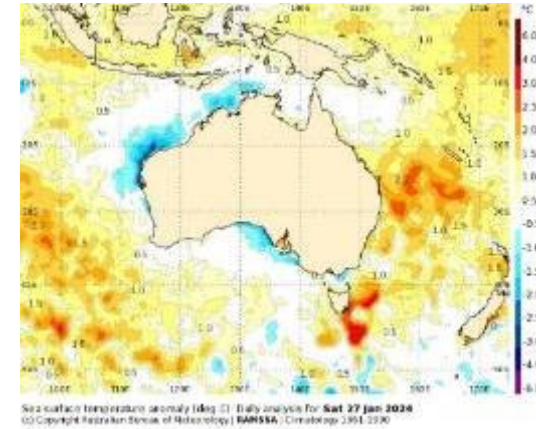
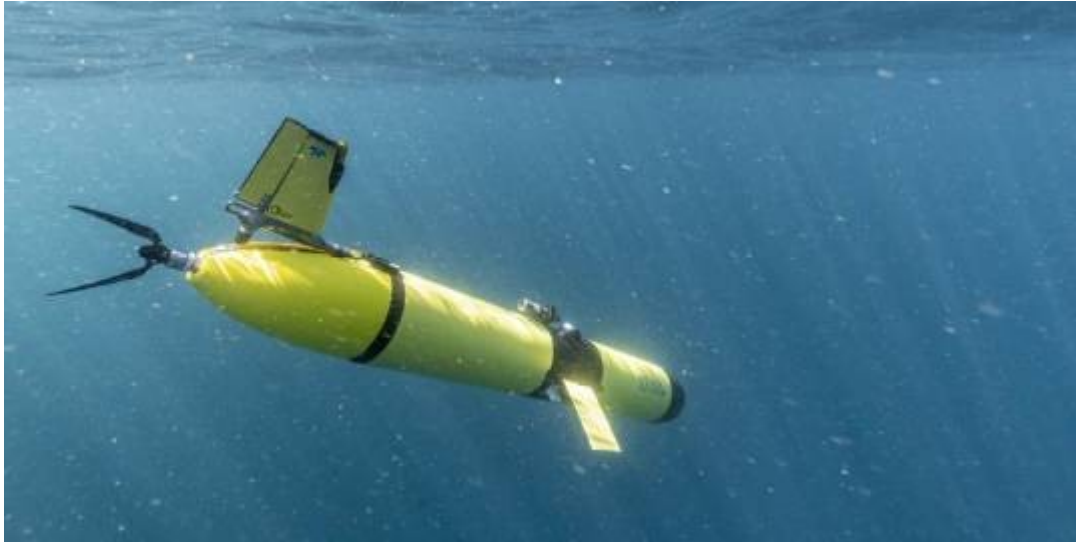
Argo Floats

IMOS is the second largest contributor to the global Argo program, which contributes real-time ocean observations to inform global weather forecasting and climate predictions.



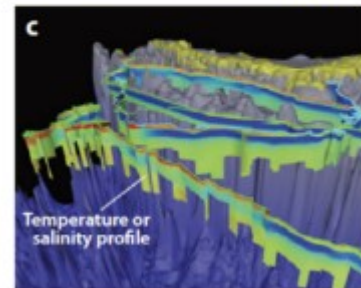
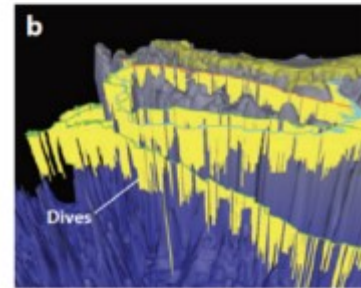
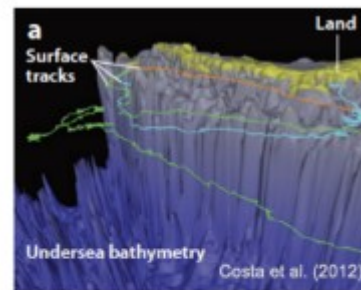
Ocean Gliders

IMOS deploys ocean gliders in shelf waters around Australia to understand sub-surface characteristics of water masses, including during extreme weather events such as marine heatwaves, tropical cyclones, river run-off, etc.



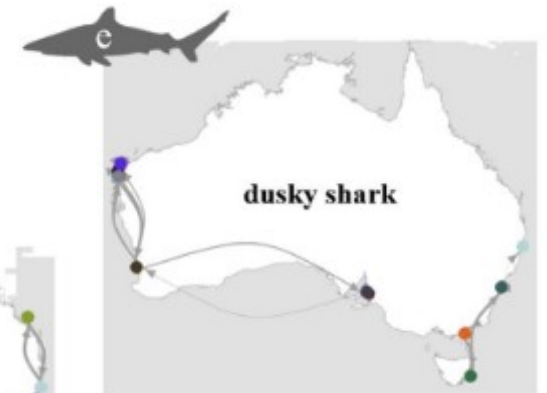
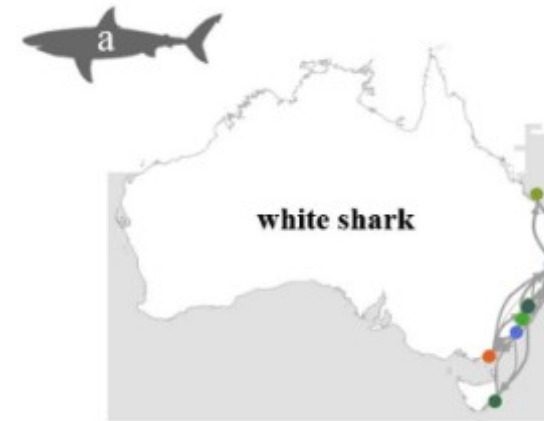
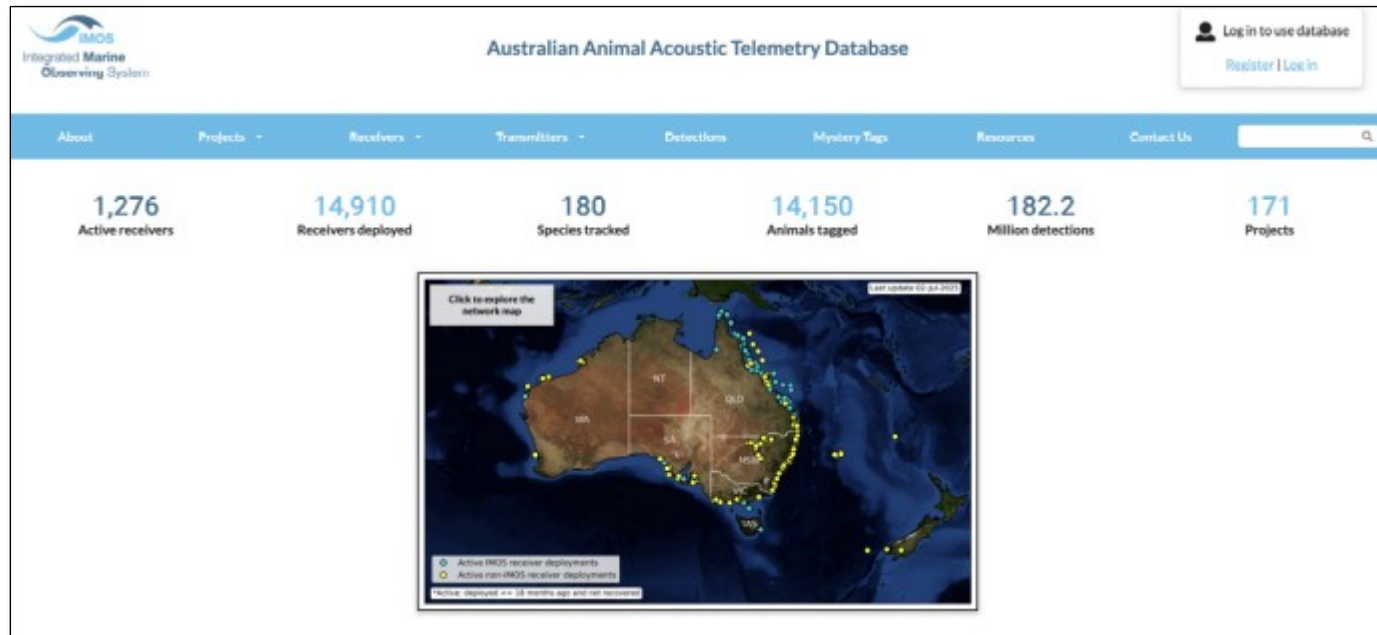
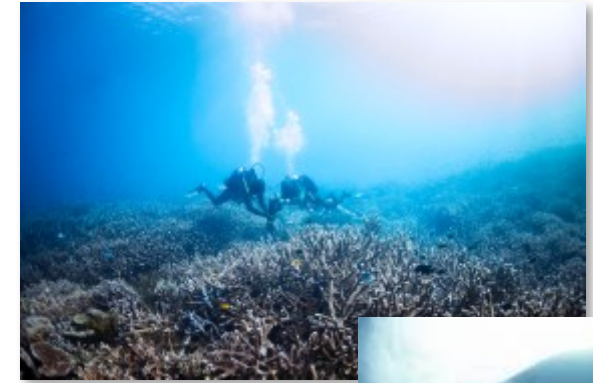
Animal-Borne Sensors

IMOS uses animals as oceanographers in places where conventional oceanographic instruments can't be deployed.

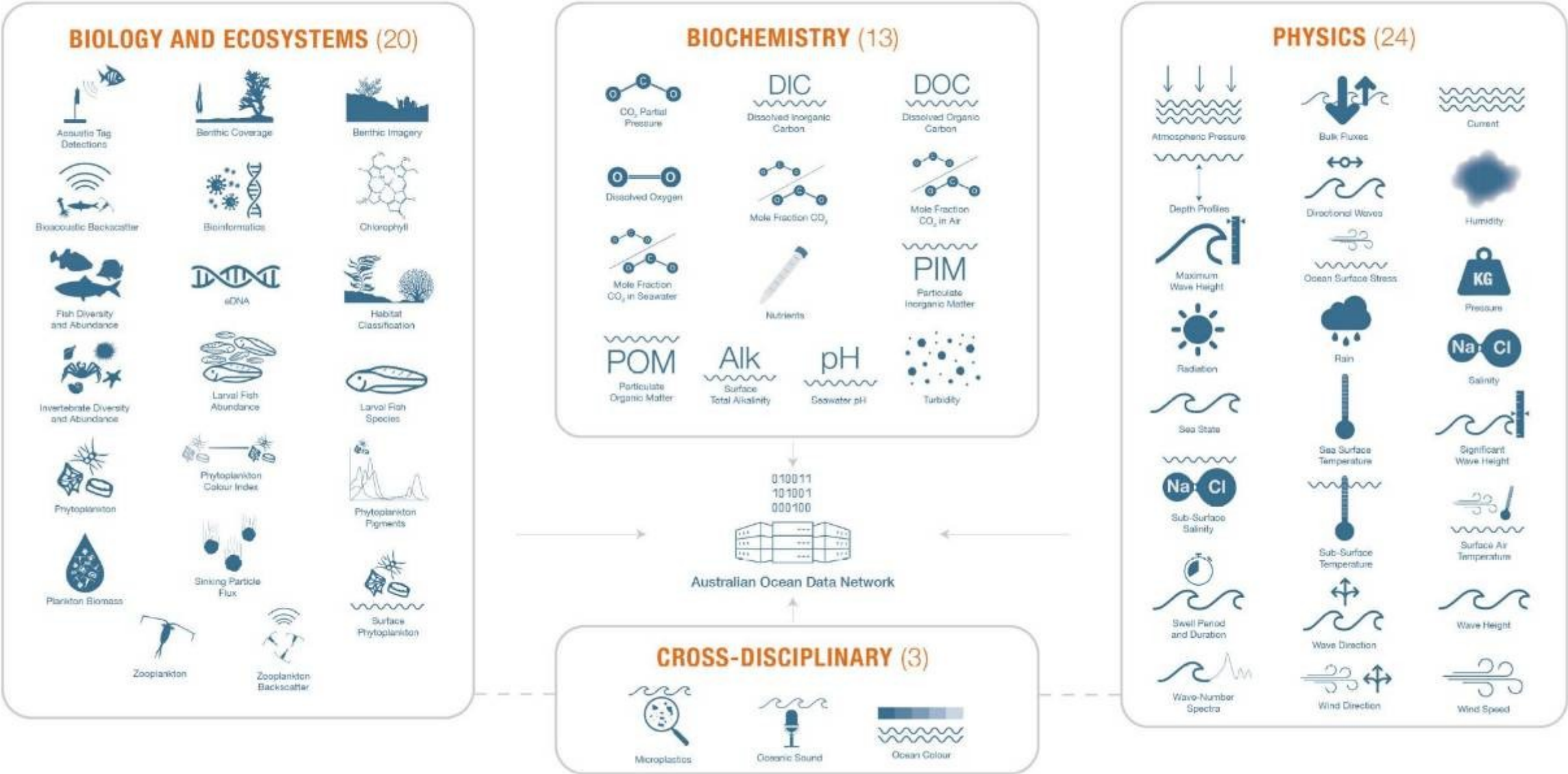


Animal Acoustic Tracking

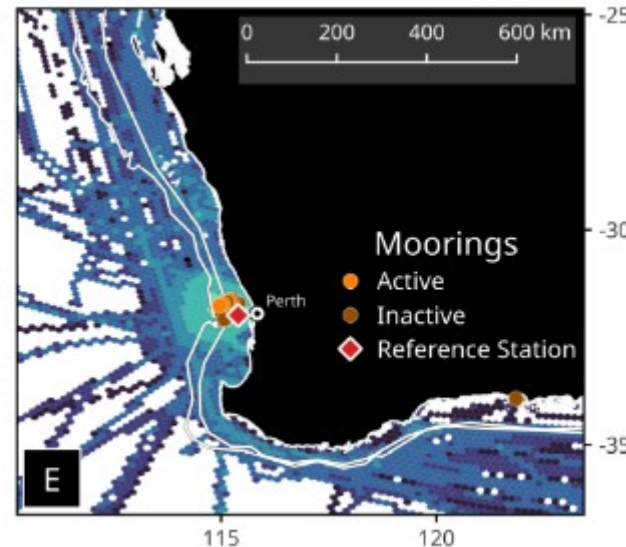
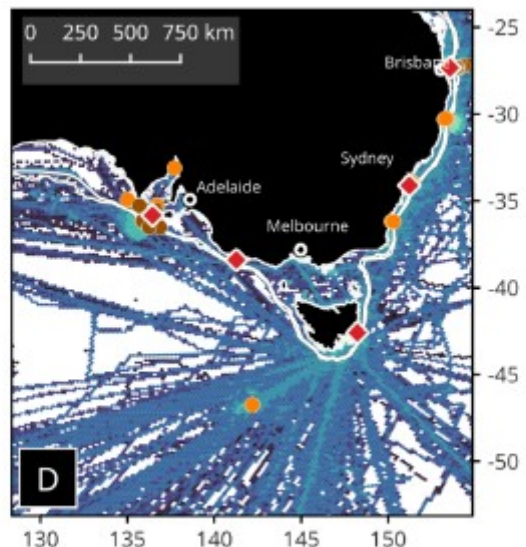
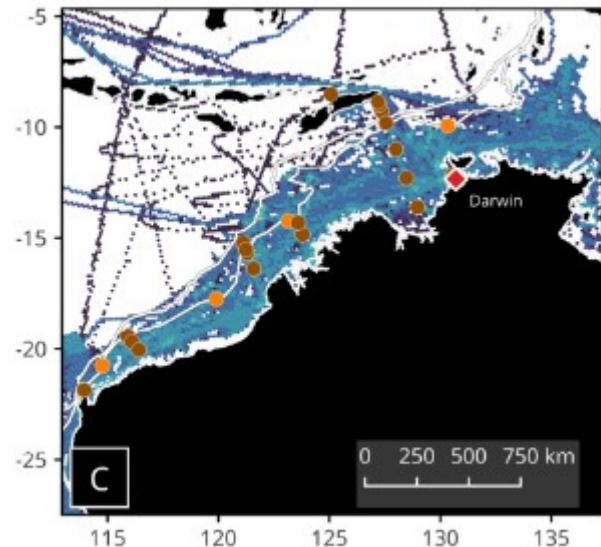
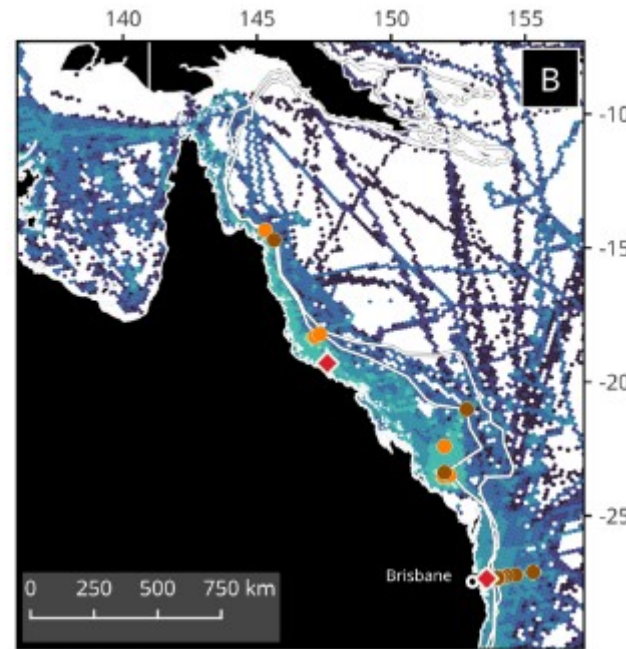
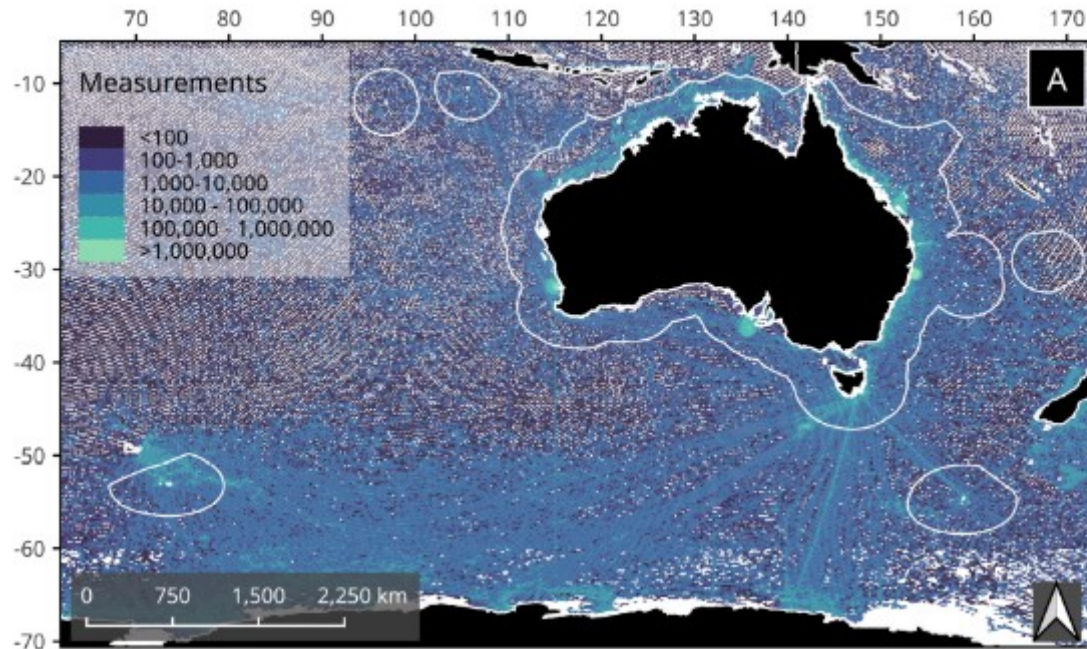
IMOS uses underwater hydrophones to detect broad-scale movements and connectivity of aquatic animals equipped with acoustic transmitters.



IMOS delivers over 60 data streams contributing to 27 Essential Ocean Variables and 21 Essential Climate Variables



The IMOS data



- +14,000 metadata records
- + 880 IMOS records
- + 180 cloud-optimised datasets
- + 1,400 variables
- + 5.7 BILLION observations

Portal - discovery

OceanCurrent - maps

IMOSlive - NRT data

BBO - Biological Ocean Observer

Remora - R pkg animal tracking

planktonR - plankton

imos-user-code library

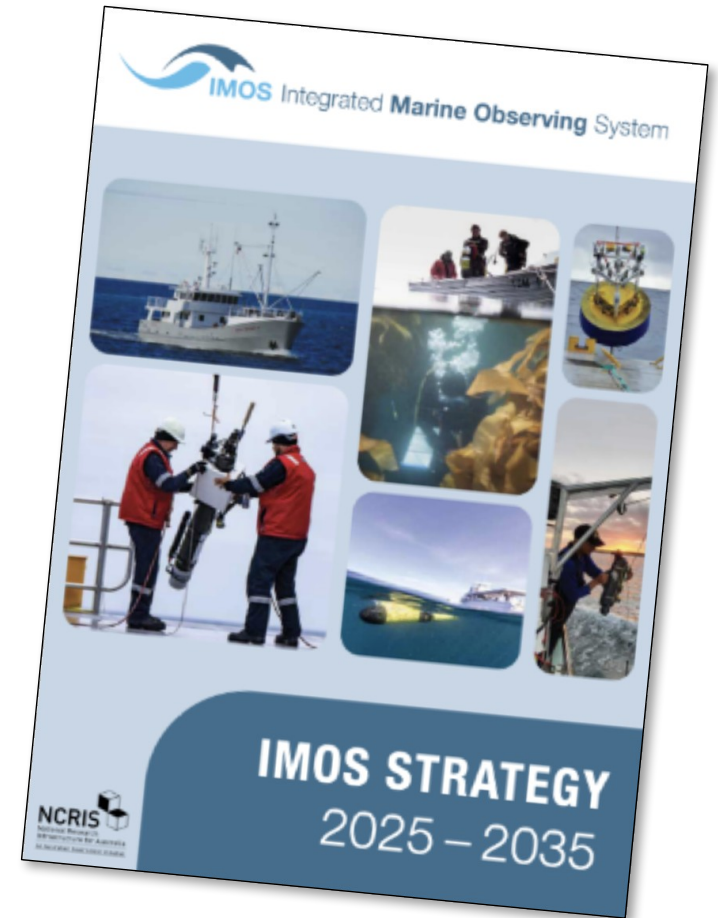
aodn cloud optimised library

How sustained are IMOS observations?

- IMOS has been funded by NCRIS since 2007 and has grown from strength to strength.
- **NCRIS funding currently secured to 2028:**
 - Continued baseline operations (11 Facilities, 47 sub-Facilities)
 - Step-change observing:
 - New (3) or enhanced observations
 - Data uplift
 - Coastal research infrastructure
- This new funding is enabling a step-change in IMOS's capacity to deliver real-time observations and data products to address stakeholder needs and inform science, operations and decision-making.

Strategic Objectives

1. **Sustain and expand IMOS' world-leading ocean observing capability and data delivery** to support healthy oceans, coastlines and communities.
2. **Empower scientific discovery and unite communities** to address the impacts of ocean-driven climate change.
3. **Provide data** to strengthen Australia's blue economy and inform industry operations, national security, and government decision-making.
4. **Achieve greater impact** through partnerships with First Nations, industry, and others.
5. **Play a leadership and stewardship role** in regional and global ocean observing.
6. Through integration and collaboration, **produce data and data products to promote a better ocean future.**



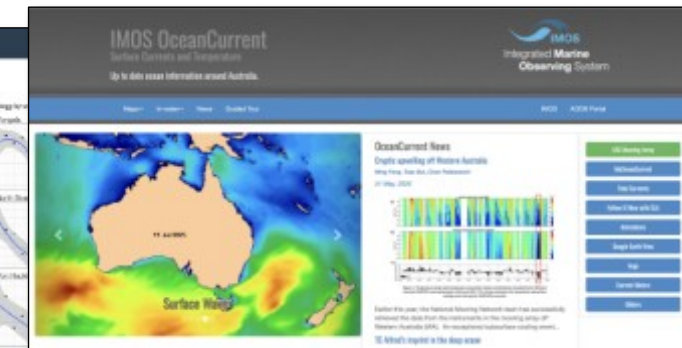
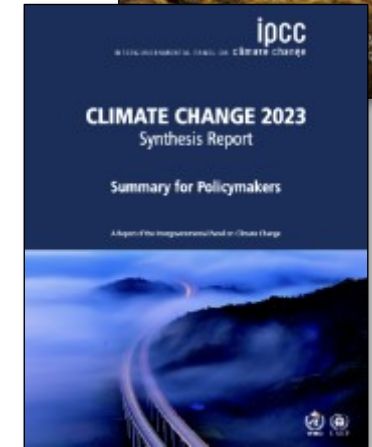
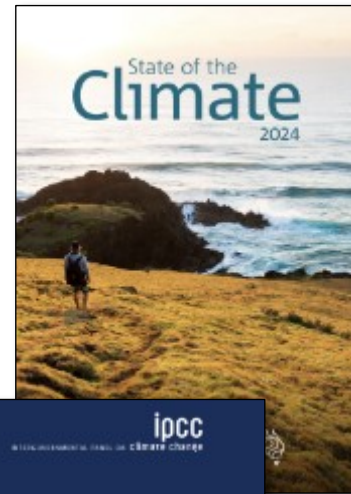
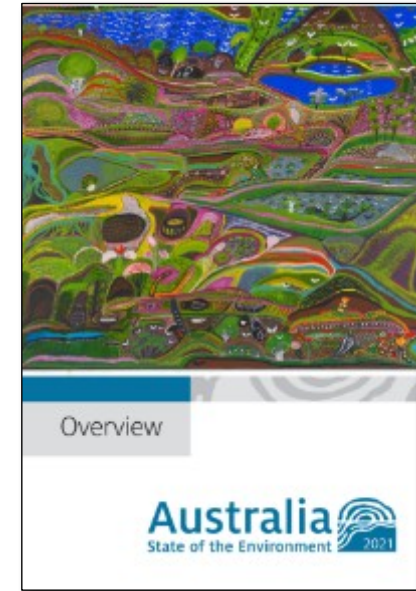
STATEMENT OF PURPOSE



Supporting solutions to environmental, climate, social and economic challenges at national and global scales through world-leading, sustained marine and Sea Country observing.

IMOS Data Uses

- IMOS data have played an important role in Australia's State of the Environment and State of the Climate reporting, Intergovernmental Panel on Climate Change projections and report and numerous policy and management documents.
- IMOS observations contribute to a range of national ocean and climate models providing forecasting services for industry operations, defence capability and extreme weather preparedness.
- IMOS data products help deliver information to decision makers.





Australia's Integrated Marine Observing System is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.

PRINCIPAL PARTICIPANTS



SIMS is a partnership involving four universities

ASSOCIATE PARTICIPANTS



IMOS thanks the many other organisations who partner with IMOS, providing co-investment, funding and operational support, including investment from the Tasmanian, Western Australian and Queensland State Governments.

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